Supplementary Material

I. Datasets

TABLE I: Group II - Number of samples and attributes for both binary (w/ IR) and multi-class (w/ number of classes) datasets

Dataset	#Samples	#Features	Imbalance Ratio
road-safety	111762	32	1.0
vehicle_sensIT	98528	100	1.0
Run_or_walk	88588	6	1.00
Dota2	102944	116	1.12
higgs_small	98049	28	1.12
phishing	11055	30	1.26
BayesianNetworkGenerator_vote	131072	16	1.59
magic_telescope	19020	11	1.84
tic-tac-toe	39366	9	1.88
codrna	488565	8	2.0
house_8L	22784	8	2.38
nomao	34465	118	2.50
adult	48842	14	3.18
skin-segmentation	245057	3	3.82
click	39948	11	4.94
bank	45211	16	7.55
santander	200000	200	8.95
giveMeCredit	150000	10	13.96
amazon	32769	9	16.27
Clickpredictionsmall	149639	11	21.37
20_newsgroups	149639	11	21.38
Image_testing	134731	30	23.14
letter	20000	16	23.60
creditcard	284807	30	577.88

Dataset	#Samples	#Features	#Classes
cmc	55296	9	3
characters	10218	7	10
endgame	44819	6	3
connect-4	67557	42	3
solar-flare	1000000	12	6
kropt	28056	6	18
shuttle	57999	9	7
fars	100968	29	8
pokerhand-normalized	829198	10	9
poker	1025009	10	10

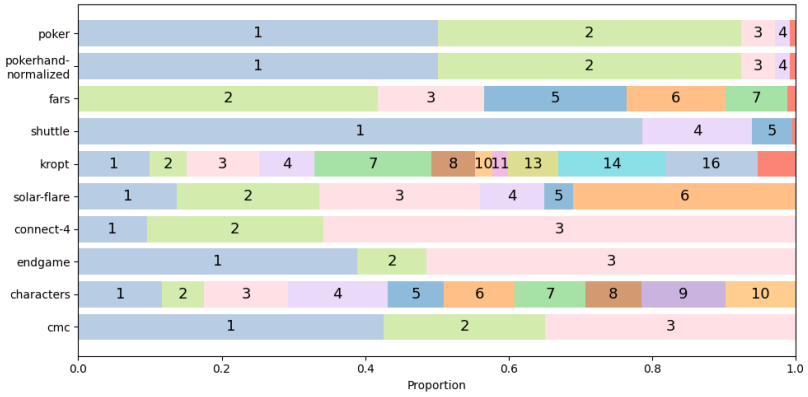


Fig. 1: Proportion of samples per class for each multiclass dataset used in this study. Each bar corresponds to a distinct multiclass dataset, with each section representing a specific class. The size of each section reflects the percentage of samples belonging to the class indicated in the label. The classes with less than 2% of samples were aggregated in the red section.

TABLE II: Number of samples, features, and Imbalance Ratio of group I datasets (Group I).

Dataset	#samples	#features	IR
prnn_synth	250	2	1
leukemia	100	50	1.04
mammographic	830	5	1.06
mammographic-cat	830	2	1.06

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Dataset	#samples	#features	IR
real-estate	414	5	1.07
solvent	52	8	1.08
Edu-Data-HvsL	269	16	1.12
Edu-Data-HvsL-cat	269	12	1.12
cryotherapy	90	6	1.14
cryotherapy-cat	90	2	1.14
sonar	208	60	1.14
redwine-2c	1599	11	1.15
schizo	112	13	1.15
schizo-cat	112	2	1.15
somerville	143	6	1.17
cleveland	297	13	1.17
cleveland-cat	297	7	1.17
wine-1vs2	130	13	1.2
irish-cat	468	3	1.22
irish	468	5	1.22
bc-coimbra	116	9	1.23
banknote	1372	4	1.25
heart-statlog	270	13	1.25
heart-statlog-cat	270	6	1.25
glioma16	50	16	1.27
lymphography-v1-cat	142	15	1.33
lymphography-v1	142	18	1.33
student-g	131	21	1.34
student-cg-cat	131	21	1.34
student-g-cat	131	10	1.34
caesarian	80	5	1.35
caesarian-cat	80	3	1.35
pharynx-1year	193	9	1.35
pharynx-1year-cat	193	6	1.35
plasma-retinol	315	13	1.37
plasma-retinol-cat	315	3	1.37
liver-disorders	345	6	1.38
traffic-cat	135	9	1.41
traffic	135	17	1.41
lupus	87	3	1.49
Edu-Data-HvsM	353	16	1.49
Edu-Data-HvsM-cat	353	12	1.49
pbc	276	17	1.49
pbc-cat	276	6	1.49
alzheimer-v1	317	9	1.5
alzheimer-v1-cat	317	2	1.5
broadwaymult0-cat	267	3	1.52
broadwaymult0	267	6	1.52
hepato-PHvsALD	294	9	1.53
student-mat-cat	395	21	1.58
student-mat	395	30	1.58
Edu-Data-MvsL	338	16	1.66
Edu-Data-MvsL-cat	338	12	1.66
audit	775	8	1.68
wdbc	569	30	1.68
bands	365	19	1.7
sports	1000	59	1.74

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Dataset	#samples	#features	IR
ionosphere	351	33	1.79
glass1	214	9	1.82
colon32	62	32	1.82
ecoli_0_vs_1	220	7	1.86
wisconsin	683	9	1.86
pima	768	8	1.87
steel-plates-faults	1941	33	1.88
iris0	150	4	2
vertebral-N	310	6	2.1
veteran-cat	137	3	2.19
veteran	137	6	2.19
thyroid-v1-cat	534	15	2.22
thyroid-v1	534	21	2.22
forest-d	523	27	2.29
diu-cat	322	4	2.32
relax	182	12	2.5
pharynx-status-cat	193	6	2.64
pharynx-status	193	9	2.64
kidney	158	24	2.67
kidney-cat	158	13	2.67
creditscore-cat	100	2	2.7
creditscore	100	6	2.7
haberman	306	3	2.78
user-know-H	403	5	2.95
gss-vw-cat	400	3	3
gss-vw	400	5	3
wifi1	2000	7	3
parkinson	195	22	3.06
transfusion	748	4	3.2
wpbc	198	32	3.21
vehicle0	846	18	3.25
tourism0	362	8	3.26
tourism0-cat	362	4	3.26
servo-cat	167	2	3.39
servo	167	4	3.39
cyyoung	189	8	3.4
cyyoung-cat	189	2	3.4
student-p-cat	131	10	3.68
student-p	131	21	3.68
immunotherapy	90	7	3.74
immunotherapy-cat	90	2	3.74
spectf	267	44	3.85
student-por-cat	649	21	3.95
student-por	649	30	3.95
icu	200	19	4
icu-cat	200	16	4
appendicitis	106	7	4.05
breast-car	106	9	4.05
dmft-mouth-cat	797	2	4.14
dmft-mouth	797	4	4.14
dmft-diet-cat	797	2	5.04
dmft-diet	797	4	5.04
newthyroid1	215	5	5.14

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Dataset	#samples	#features	IR
hepatitis-cat	80	13	5.15
hepatitis	80	19	5.15
dmft-all-cat	797	2	5.28
dmft-all	797	4	5.28
dmft-health-cat	797	2	5.43
dmft-health	797	4	5.43
thoracic	470	16	5.71
thoracic-cat	470	13	5.71
forest-fires	517	12	5.8
forest-fires-cat	517	2	5.8
segment0	2308	19	6.02
broadway2	89	8	7.09
broadway2-cat	89	3	7.09
broadway3	89	8	7.09
fertility-diagnosis	100	9	7.33
fertility-diagnosis-cat	100	7	7.33
pharynx-3year	193	9	7.77
pharynx-3year-cat	193	6	7.77
broadwaymult6	267	6	8.54
diu-ro10-cat	322	5	9.06
broadwaymult4	267	6	9.68
vowel0	988	13	9.98
ctg-pathologic	2126	21	11.08
broadwaymult3-cat	267	3	11.71
broadwaymult3	267	6	11.71
broadwaymult5	267	6	11.71
tourism-23457vs01	362	8	12.41
tourism-23457vs01-cat	362	4	12.41
shuttle_c0_vs_c4	1829	9	13.87
page_blocks_1_3_vs_4	472	10	15.86
dermatology6	358	34	16.9
lymphography-normal-fibrosis-cat	148	15	23.67
lymphography-normal-fibrosis	148	18	23.67
tourism2	362	8	26.85
tourism2-cat	362	4	26.85
winequality_red_4	1599	11	29.17
poker_9_vs_7	244	10	29.5
diu-bs10-cat	322	5	31.2
ecoli-0-1-3-7_vs_2-6	281	7	39.14
winequality-white-3_vs_7	900	11	44

II. Statistical Results

TABLE III: Wilcoxon Test for the for the Group I binary datasets. All values are below 0.05, showing that FPDBSCV with threshold tuning significantly outperforms DBSCV.

Classifier	p-value	< 0.05
NB	.00008	Yes
NN1	.00001	Yes
NN3	.00001	Yes
Random Forest	.00001	Yes
XG-Boost	.00001	Yes

TABLE IV: Wilcoxon test for the Group II binary datasets. All values, excpet for NB classifier are below 0.05, showing that FPDBSCV with threshold tuning significantly outperforms SCV.

Classifier	p-value	Significant (< 0.05)
NB	0.4777	No
NN1	0.00244	Yes
NN3	0.00036	Yes
RF	0.0001	Yes
XGB	0.0014	Yes